

Factory Reallocation and Shipping Optimization Using Data-Driven Decision Intelligence: A Case Study of Nassau Candy Distributor

| Submitted By : Ms.Prerana Sachin Khedekar
| Internship Project

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Abstract

Efficient supply-chain management requires organizations to balance delivery speed, operational efficiency, profitability, and risk. In distribution-intensive businesses, factory allocation and shipping decisions can significantly influence customer service levels and overall operational performance. This project presents a data-driven decision-intelligence framework for evaluating factory reallocation and shipping optimization scenarios for Nassau Candy Distributor.

The proposed system analyzes historical order information together with simulated factory scenarios to identify potentially more efficient factory assignments for specific product, region, and shipping-mode combinations. The system evaluates current and scenario-based lead times, lead-time reduction, optimization scores, confidence levels, risk classifications, profit margins, order volumes, and units handled. A Streamlit-based interactive dashboard was developed to make the analytical results accessible to decision-makers.

The dashboard provides six major analytical capabilities: factory optimization simulation, what-if scenario analysis, recommendation ranking, risk and confidence assessment, business-impact analysis, and project methodology/data exploration. Users can dynamically select a product, geographical region, and shipping mode and adjust the relative priority between speed and profit. The system then presents the relevant factory scenarios and identifies recommendations using optimization scores.

An important methodological consideration is that the available source dataset does not contain observed shipments of the same product from multiple factories. Consequently, the factory recommendations represent analytical simulations rather than direct observations of real-world alternative factory assignments. The proposed framework should therefore be considered a decision-support prototype and its recommendations should be operationally validated before implementation.

Keywords: Supply Chain Analytics, Factory Allocation, Shipping Optimization, Lead Time, Decision Intelligence, What-If Analysis, Streamlit, Data Analytics, Risk Analysis, Business Intelligence

1. Introduction

1.1 Background

Modern supply chains operate in an environment where organizations must continuously balance customer expectations, transportation requirements, operational constraints, and profitability. One important factor affecting supply-chain performance is the time required to fulfill and deliver customer orders.

For a distribution organization such as Nassau Candy Distributor, the selection of an appropriate factory for a particular product, region, and shipping mode can influence delivery lead time and operational efficiency. A factory allocation that appears suitable from a traditional operational perspective may not always provide the shortest possible lead time for every product-region combination.

Data analytics provides an opportunity to examine historical performance and evaluate alternative scenarios before making operational decisions. Instead of relying exclusively on intuition, organizations can use historical data, predictive measures, optimization scores, and scenario analysis to support more informed decisions.

This project develops a decision-intelligence dashboard for evaluating factory reallocation and shipping optimization scenarios. The application uses historical order data, factory recommendations, and scenario results as analytical inputs. The Streamlit application loads three primary datasets: historical cleaned data, factory recommendations, and scenario results.

1.2 Problem Statement

The primary problem addressed by this project is the identification of potentially better factory assignments that can reduce shipping lead time while considering profitability, optimization performance, confidence, and risk.

Traditional analysis may make it difficult to compare multiple factory alternatives for different products, regions, and shipping modes. Furthermore, decision-makers may require an interactive mechanism to evaluate different priorities, such as whether delivery speed should receive greater importance than profitability.

Therefore, this project proposes an interactive analytical framework that:

1. Examines historical shipping performance.
2. Evaluates simulated factory alternatives.
3. Estimates scenario-based lead times.
4. Measures potential lead-time reductions.
5. Ranks alternatives using optimization scores.
6. Provides confidence and risk indicators.
7. Supports what-if analysis.
8. Presents recommendations through an interactive dashboard.

1.3 Research Objective

The main objective of this research is to develop a data-driven decision-support system for factory reallocation and shipping optimization.

The specific objectives are:

- To analyze historical order and shipping information.
- To evaluate factory alternatives for different product-region-shipping combinations.
- To compare current and recommended factory lead times.
- To calculate potential lead-time reductions.
- To rank factory scenarios using optimization scores.
- To incorporate scenario confidence and risk into decision-making.
- To provide interactive what-if analysis.
- To summarize potential business impact through key performance indicators.
- To develop an accessible Streamlit dashboard for operational decision support.

1.4 Research Questions

This study focuses on the following research questions:

RQ1: Can historical shipping data be used to identify potentially better factory assignments?

RQ2: How can alternative factory scenarios be compared using predicted lead time?

RQ3: How can lead-time reduction and optimization scores support factory allocation decisions?

RQ4: How can confidence and risk indicators improve the interpretation of analytical recommendations?

RQ5: Can an interactive dashboard improve the accessibility and usability of supply-chain optimization results?

2. Literature and Conceptual Background

2.1 Supply Chain Optimization

Supply-chain optimization involves selecting operational decisions that improve performance while considering multiple constraints and objectives. Common objectives include reducing transportation time, controlling cost, improving customer service, and maintaining profitability.

Factory allocation is one component of supply-chain optimization because the physical location and assignment of manufacturing or distribution facilities can affect transportation and fulfillment performance.

2.2 Lead-Time Optimization

Lead time represents the duration associated with fulfilling or delivering an order. Reducing lead time can potentially improve customer satisfaction and supply-chain responsiveness.

In this project, current lead time is compared with scenario lead time. The difference between these values represents the potential time saving associated with a simulated factory reallocation.

The application explicitly calculates the difference between current and recommended lead time and displays it as "Time Saved."

2.3 Decision Intelligence

Decision intelligence combines data, analytical techniques, scenario evaluation, and visualization to support business decisions.

Rather than presenting only historical descriptive statistics, this project provides scenario-based recommendations. Users can compare current and recommended factories and examine their respective lead times.

2.4 What-If Analysis

What-if analysis allows decision-makers to examine the possible effects of alternative choices before implementing them.

In this system, users can select a product, region, and shipping mode and examine available factory scenarios. The application also allows users to adjust speed priority from 0 to 100, with profit priority automatically calculated as the remaining percentage.

3. Proposed Methodology

3.1 Research Framework

The proposed framework consists of the following stages:

Historical Data → Data Cleaning → Performance Analysis → Factory Scenario Generation → Lead-Time Evaluation → Optimization Scoring → Confidence/Risk Assessment → Recommendation → Interactive Dashboard

This workflow transforms raw analytical information into decision-oriented recommendations.

3.2 Data Sources

The application expects three principal CSV datasets:

- Dataset 1: Historical Order Data

The historical dataset is loaded from:

"nassau_candy_cleaned.csv"

It represents the historical information used as the foundation for analysis.

- Dataset 2: Factory Recommendations

The recommendation dataset is loaded from:

"factory_recommendations.csv"

It contains candidate factory recommendations and associated analytical measures.

- Dataset 3: Scenario Results

The scenario dataset is loaded from:

"scenario_results.csv"

It contains alternative factory scenarios and their predicted performance.

The application verifies whether all three files are available before continuing with the analysis.

3.3 Data Preparation

Before analysis, column names are standardized by removing unnecessary spaces. Text-based fields such as product name, region, shipping mode, current factory, scenario factory, scenario confidence, and risk are cleaned and converted to string format.

Numerical variables including current lead time, scenario lead time, lead-time reduction, lead-time reduction percentage, optimization score, confidence score, profit margin, orders, units, sales, gross profit, and cost are converted into numerical formats.

This preprocessing improves consistency and reduces the possibility of data-type-related errors during analysis.

3.4 Product-Region-Shipping Mode Segmentation

The application creates valid combinations of:

- Product
- Region
- Ship Mode

Only combinations available in the recommendation data are presented to the user. This prevents invalid combinations from appearing in the dashboard.

This segmentation allows the analysis to become more specific and operationally meaningful.

3.5 Factory Scenario Evaluation

For every selected product-region-shipping combination, the system identifies available factory scenarios.

The scenario analysis considers:

- Current factory
- Scenario factory
- Current lead time
- Scenario lead time
- Lead-time reduction
- Lead-time reduction percentage
- Optimization score
- Scenario confidence

- Risk

The available scenarios are presented in an interactive table and visually compared using a factory-versus-predicted-lead-time chart.

3.6 Optimization Score

The optimization score is used to rank available factory alternatives.

The system selects the recommendation with the highest optimization score for the selected product, region, and shipping mode. The same principle is applied when identifying the overall best recommendation.

Because the exact mathematical formulation of the optimization score is not contained in the uploaded Streamlit application, this paper does not claim a specific mathematical formula for that score.

3.7 Lead-Time Reduction

Lead-time reduction is calculated by comparing the current lead time with the scenario lead time.

Conceptually:

Lead-Time Reduction = Current Lead Time – Scenario Lead Time

The percentage reduction is provided by the recommendation dataset and displayed as a key performance indicator.

The dashboard presents:

- Current Lead Time
- Recommended Lead Time

- Lead-Time Reduction %
- Scenario Confidence

as the primary KPIs for a selected scenario.

3.8 Confidence and Risk Analysis

Analytical recommendations may not have equal levels of reliability. Therefore, the dashboard categorizes recommendations according to scenario confidence.

The application counts:

- High-confidence recommendations
- Medium-confidence recommendations
- Low-confidence recommendations
- Total recommendations

It also provides a risk distribution when risk information is available.

This provides decision-makers with additional context rather than presenting recommendations without uncertainty information.

4. System Architecture

4.1 Technology Stack

The project uses the following technologies:

Technology| Purpose

Python| Core programming and data processing

Pandas| Data loading, cleaning, filtering and analysis

NumPy| Numerical processing

Streamlit| Interactive dashboard development

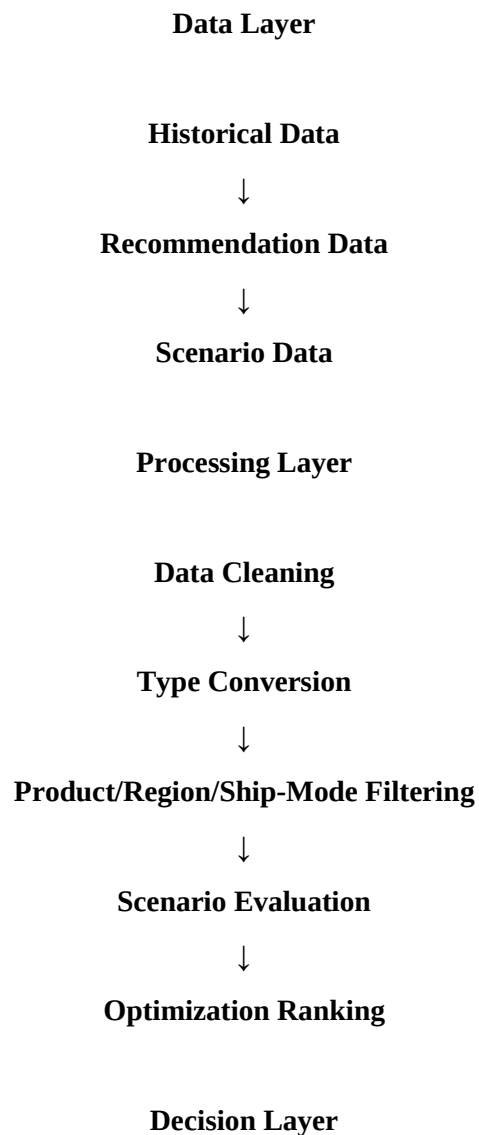
Pathlib| File and project-path management

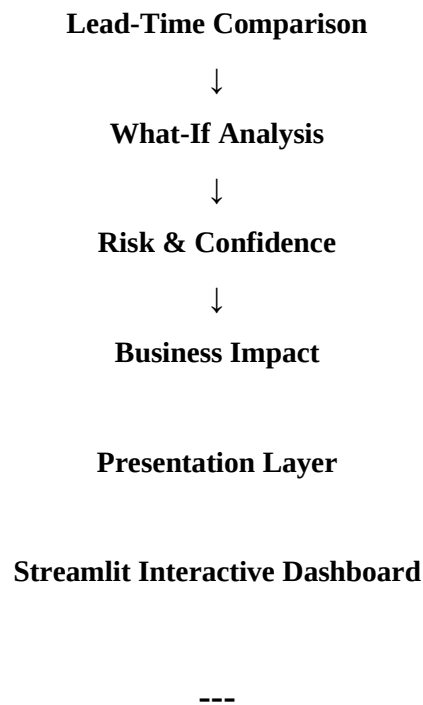
CSV| Data storage and exchange

The application imports Streamlit, Pandas, NumPy, and Pathlib and uses Streamlit to construct the dashboard interface.

4.2 Application Architecture

The application can be represented as:





5. Dashboard Design and Functional Modules

5.1 Factory Optimization Simulator

The first major module displays available factory scenarios for the selected product, region, and shipping mode.

The dashboard provides a scenario table containing factory, scenario factory, lead-time values, reduction, optimization score, confidence, and risk information.

5.2 What-If Scenario Analysis

The second module compares the current factory against the recommended factory.

The comparison includes:

- Current factory
- Recommended factory

- Current lead time
- Recommended lead time
- Time saved

This allows decision-makers to understand the operational effect of a potential factory change.

5.3 Recommendation Dashboard

The recommendation module ranks the top factory recommendations according to optimization score.

The dashboard displays the top 10 recommendations and identifies the highest-scoring recommendation as the "Best Overall Recommendation."

5.4 Risk and Impact Panel

The risk panel provides a high-level view of recommendation confidence and risk.

The dashboard visualizes confidence distribution and, when available, risk distribution. This supports more cautious interpretation of analytical recommendations.

5.5 Business Impact

The business-impact module calculates:

- Average lead-time reduction
- Average profit margin
- Total orders analyzed
- Total units analyzed

These indicators help connect the analytical model to business performance.

5.6 Data Exploration

The application also provides expandable sections for viewing:

- Historical dataset
- Factory recommendations
- Scenario results

This improves transparency by allowing users to inspect the underlying analytical information.

6. Results and Analytical Interpretation

The system is designed to generate several categories of analytical results.

6.1 Factory-Level Results

For each valid product-region-shipping combination, alternative factory assignments can be compared according to predicted lead time.

A lower scenario lead time indicates a potentially faster fulfillment configuration.

6.2 Recommendation Results

The highest optimization score is used to identify the preferred recommendation within the available scenario records.

The system also produces an overall best recommendation by sorting all recommendation records according to optimization score.

6.3 Operational Impact

The system measures the difference between current and recommended lead times. This provides a direct indicator of potential time savings.

6.4 Business Impact

The dashboard summarizes average lead-time reduction, profit margin, total orders, and total units analyzed.

These indicators can be used by managers to understand the scale and potential importance of the analytical recommendations.

7. Discussion

The developed system demonstrates how supply-chain data can be transformed into an interactive decision-support environment.

One of the key advantages of the proposed framework is that it does not restrict analysis to historical reporting. Instead, it introduces scenario-based evaluation, allowing users to compare alternative factory assignments.

The interactive selection of product, region, and shipping mode makes the analysis more granular. A decision-maker can focus on a particular operational combination rather than examining the entire dataset at once.

Another important feature is the ability to consider different decision priorities. The dashboard provides a speed-priority slider ranging from 0 to 100, while profit priority is calculated as the complementary percentage.

This reflects a practical business reality: an organization may sometimes prioritize faster delivery, while in other circumstances profitability may be more important.

The inclusion of confidence and risk indicators is also valuable. Optimization recommendations should not be treated as equally reliable, particularly when they are generated from simulated scenarios. The dashboard therefore provides confidence classifications and risk distributions to support more informed interpretation.

8. Limitations

The project has several important limitations.

8.1 Simulated Factory Scenarios

The most significant limitation is that the source dataset does not contain observed shipments of the same product from multiple factories. Therefore, the alternative factory results are analytical simulations rather than direct observations.

The application itself explicitly states that factory recommendations should be validated operationally before implementation.

8.2 Dependence on Historical Data

The quality of analytical recommendations depends on the quality and representativeness of the historical data.

Changes in demand patterns, transportation conditions, factory capacity, or operational policies may affect future performance.

8.3 Scenario Assumptions

Scenario-based lead times depend on the assumptions used to generate the scenario results. The Streamlit application consumes scenario results but does not expose the complete mathematical derivation of those values.

Therefore, this research does not claim that the predicted lead times are guaranteed real-world outcomes.

8.4 Optimization Score Transparency

The uploaded dashboard uses an optimization score to rank recommendations, but the exact formula used to generate the score is not included in the Streamlit application.

Future versions should document the mathematical formulation and weighting methodology used to calculate the score.

9. Future Scope

The project can be enhanced in several ways.

9.1 Real-Time Data Integration

The dashboard could be connected to live ERP, warehouse-management, transportation-management, or order-management systems.

9.2 Machine Learning-Based Lead-Time Prediction

Machine-learning models could be introduced to predict delivery lead time using factors such as product characteristics, region, shipping mode, historical delivery patterns, seasonality, and operational conditions.

9.3 Cost Optimization

Future versions could explicitly optimize both transportation cost and lead time.

9.4 Multi-Objective Optimization

A more advanced optimization model could simultaneously consider:

- Lead time
- Transportation cost
- Profitability
- Factory capacity
- Customer priority

- Risk
- Service-level requirements

9.5 Capacity Constraints

Factory capacity could be incorporated so that recommendations do not assign unrealistic volumes to a factory.

9.6 Advanced Visualization

Future dashboards could include interactive maps, trend analysis, heatmaps, scenario comparison charts, and executive-level summary pages.

9.7 Model Validation

The most important future improvement would be validation using real-world shipments from multiple factories for comparable products. This would allow simulated recommendations to be compared against observed operational outcomes.

10. Conclusion

This project presents a data-driven framework for factory reallocation and shipping optimization for Nassau Candy Distributor. The developed Streamlit dashboard transforms historical data and simulated factory scenarios into an interactive decision-support system.

The framework evaluates product-region-shipping combinations, compares current and alternative factories, measures potential lead-time reduction, ranks recommendations using optimization scores, and provides confidence and risk information. It also includes what-if analysis and business-impact indicators, enabling users to examine potential operational improvements from multiple perspectives.

The project demonstrates the value of combining data analytics, scenario analysis, optimization concepts, and interactive visualization in supply-chain decision-making.

However, the recommendations should be interpreted as analytical simulations rather than confirmed operational outcomes. Since the underlying dataset does not contain observed shipments of the same product from multiple factories, operational validation is necessary before implementing any factory-reallocation recommendation.

Overall, the proposed system provides a strong foundation for developing a more advanced supply-chain decision-intelligence platform capable of supporting factory allocation, delivery-time optimization, risk evaluation, and strategic operational planning.

References

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Appendix A – Key Dashboard Components

The developed application contains the following major sections:

1. Key Performance Indicators
2. Factory Optimization Simulator
3. What-If Scenario Analysis
4. Recommendation Dashboard
5. Risk & Impact Panel
6. Business Impact
7. About Project and Methodology
8. Dataset Information
9. Historical Dataset Preview
10. Factory Recommendation Preview
11. Scenario Results Preview

These components collectively provide analytical, operational, and managerial views of the factory optimization problem.

Appendix B – Project Disclaimer

The factory recommendations generated by this project are analytical simulations based on historical product-region-shipping performance and assumed factory scenario factors. The source dataset does not contain observed shipments of the same product from multiple factories. Therefore, scenario results should be validated operationally before implementation.